

PRADEEP KUMAR BALASUBRAMANIAN

Embedded Software Engineer | Automotive Telematics & RTOS

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PROFESSIONAL SUMMARY

Embedded Software Engineer with 4+ years of experience building production firmware in C/C++ and Python across automotive and industrial systems. Specialized in RTOS-based real-time design, bootloader development, flash memory management, and communication protocol integration (CAN, J1939, UDS ISO 15765-4, MODBUS). Hands-on track record in automotive telematics, ECU diagnostics, FOTA workflows, and IoT device firmware.

TECHNICAL SKILLS

Programming:	Embedded C, C++, Python, Shell Scripting
Platforms & MCUs:	Renesas RA Series, Renesas RH850, ARM Cortex-M (STM32), ESP32, ESP8266, PIC, Quectel EC200
RTOS & Architecture:	FreeRTOS, QL_RTOS (QuecOpen), Custom Bootloaders, Flash Memory, Dual-Bank FOTA, ISR
Protocols & Bus:	CAN, J1939, UDS (ISO 15765-4), XCP, K-Line, J1708, UART, SPI, I2C, MODBUS, MQTT
Tools & Equipment:	Renesas e2 studio, STM32CubeIDE, Keil µVision, Logic Analyzers, Oscilloscopes, Git, ROS

WORK EXPERIENCE

Embedded Systems Engineer

Pune, India | Aug 2024 – Present

Intangles Lab Pvt Ltd

- Engineered telematics firmware and custom bootloaders on Renesas RA Series MCUs, implementing flash memory management for backup firmware images, update flags, and persistent configuration data.
- Architected Quectel EC200 (QuecOpen / QL_RTOS) firmware managing end-to-end CAN polling, ECU communication, cloud routing, and FOTA workflows.
- Integrated and validated automotive communication protocols including CAN, J1939, ISO 15765-4 (UDS), XCP, K-Line, and J1708 for vehicle diagnostics.
- Performed system-level debugging, performance optimization, and fault analysis to maximize firmware stability and communication reliability.
- Developed a Python-based automation application to orchestrate ECU FOTA validation workflows.

Embedded Engineer

Coimbatore, India | Jul 2020 – Aug 2022

Nanotronics Scientific Solutions

- Developed production-ready embedded firmware for IoT devices, industrial fuel dispensers, and home automation controllers.
- Implemented and optimized multi-protocol communication using UART, MODBUS, SPI, and MQTT for deterministic real-time data exchange.
- Integrated FreeRTOS, designing deterministic tasks, queues, semaphores, and software timers for responsive system performance.
- Supported hardware bring-up, debugging, and production deployment across ESP32, ESP8266, STM32, and PIC platforms.

Project Intern — Range-Extended EV

Bengaluru, India | Jul 2019 – Aug 2019

SAEINDIA (in collaboration with General Motors)

- Led electronics and embedded systems development for a Range-Extended Electric Vehicle (REEV) project with General Motors.
- Engineered embedded control logic, sensor interfacing, and vehicle performance validation; awarded Best Performance & Runner-up nationally.

EDUCATION & PUBLICATION

Postgraduate Certificate, Intelligent Robotics

University of York, United Kingdom (Sep 2022 – Dec 2023)

Key Projects: Autonomous Robotic Car, Robotic Arm manipulation, Rescue Robot with ROS

B.E., Electronics & Communication Engineering

Sri Shakthi Institute of Engineering & Technology, India (2016 – 2020)

Key Project: Range-Extended Electric Vehicle embedded powertrain control

Publication: *Automotive Electronic System Controlling the Range Extended Electric Vehicle — IJERT, ICEECT 2020, Vol 8, Issue 17.*